



Nagar YuwakShikshanSanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to RashtasantTukadojiMaharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	Mission: Means to achieve Vision
---	---

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

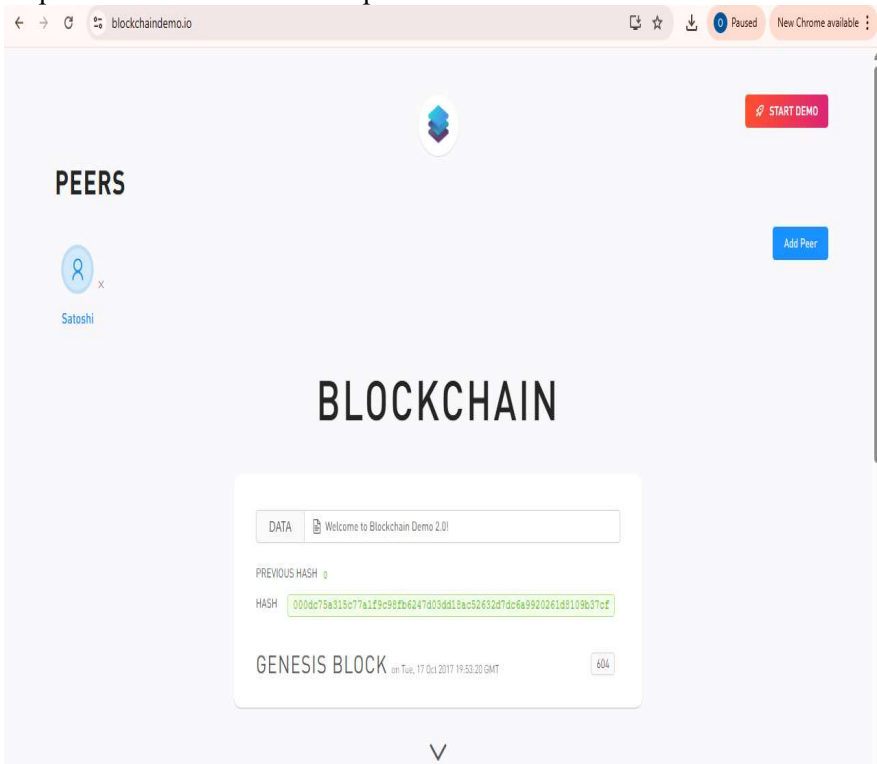
Sheikh Wasimuddin
Name of Student

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.*

Session	2026-27 (ODD)	Course Name	PE-V : Lab : Blockchain Technology
Semester	7	Course Code	23IOT1748
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	01
Course Outcome	1. Demonstrate basics of distributed systems and cryptographic primitives used in blockchain. 2. Outline various research aspects for development of blockchain. 3. Make use of different security protocols to build secure blockchain applications. 4. Analyze smart contracts and Hyperledger Fabric used in blockchain. 5. Determine the performance of applications developed with Bitcoin and Ethereum blockchain.
Aim	Introduction to blockchain technology and exploration of real world blockchain applications.
Theory (100 words)	Blockchain technology is a decentralized digital ledger that securely records transactions across a network of computers. Information is stored in blocks that are linked together using cryptographic techniques, forming a chain that is highly resistant to tampering. Unlike traditional databases controlled by a central authority, blockchain allows all authorized participants to access and verify the same data, improving transparency, security, and trust. Once a transaction is validated through a consensus mechanism, it becomes a permanent part of the blockchain, making the system reliable and difficult to manipulate. Blockchain has applications far beyond cryptocurrencies. In the financial sector, it enables secure digital payments, faster cross-border transactions, and smart contracts that automatically execute agreements when predefined conditions are met. In supply chain management, blockchain improves product traceability by recording every stage of a product's journey, helping prevent fraud and counterfeiting. Healthcare organizations use blockchain to securely store and share patient records while protecting privacy. Governments are adopting blockchain for digital identity systems, land registration, and transparent voting processes. Educational institutions also use blockchain to issue and verify academic certificates. With its ability to

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.*

Procedure and Execution (100 Words)	provide secure, transparent, and efficient data management, block chain is becoming a transformative technology across many industries and continues to shape the future of digital innovation. Implementation Steps: 1. Open the website blockchaindemo.io. 2. See the first block called the Genesis Block. 3. Type some data and add a new block. 4. Notice each block has its own hash and links to the previous one. 5. If you change the data, the hash changes and the chain breaks. 6. Use the Mine button to fix the block and make the chain valid again Now to Study the difference between two blocks: 1. Go to Blockchain explorer 2. Select Ethereum 3. Now select any two block Stepwise Screenshots with steps: 
---	---



Nagar YuwakShikshanSanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to RashttrasantTukadojiMaharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++



Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

DATA

sheikh wasimuddin

PREVIOUS HASH 000dc75a315c77a1f9c98fb6247d03dd18ac52632d7dc6a9920261d8109b37cf

HASH 0000920ae607d0d8d897f7ab565b4d46fd642ac20295f63b5d537a830fd11779

BLOCK #1 on Tue, 14 Jul 2026 06:05:06 GMT 8629

▼

DATA

CSEIOT 61

PREVIOUS HASH 0000920ae607d0d8d897f7ab565b4d46fd642ac20295f63b5d537a830fd11779

HASH 000f7babe7260850a1753fc9b31d1ed11e71dfa6f81ebc7043f606f52e6abb6e

BLOCK #2 on Tue, 14 Jul 2026 06:05:16 GMT 3228

▼



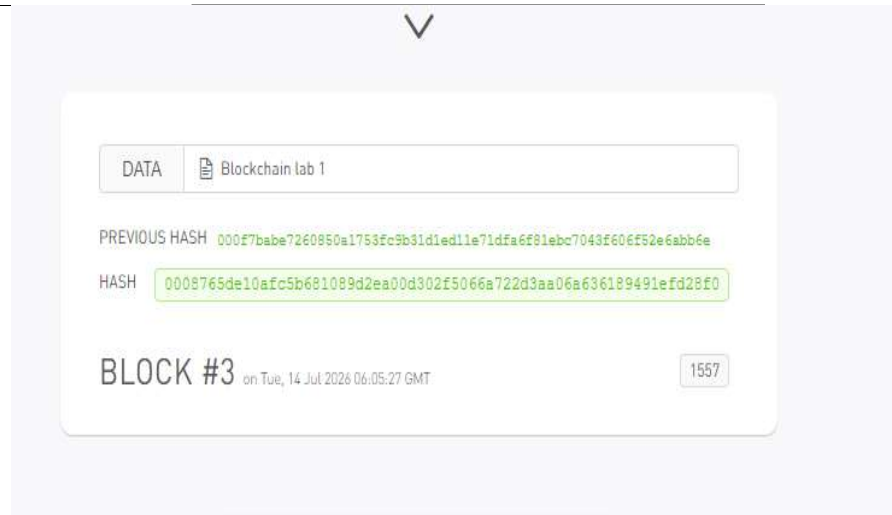
Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.



Blockchain

- Home
- Prices
- Charts
- SnapMarkets **NEW**
- NFTs
- DeFi
- Prediction Markets
- Academy
- News
- Developers
- Wallet
- Bitcoin
- Ethereum
- Solana
- Bitcoin Cash
- English

Search Blockchain, Transactions, Addresses, Prediction Markets and Blocks

Sign In

Play Gaming



Ethereum Block 25,528,987

Mined on July 14, 2026 11:37:59 • All Blocks

Unknown Miner

A total of 64,2828 ETH (\$114,708) were sent in the block with the average transaction being 0.1461 ETH (\$260.70). 0x48-5f97 earned a total reward of 0.01 ETH \$13.41. The reward consisted of a base reward of 0.01 ETH \$13.41 with an additional 0.0196 ETH (\$35.03) reward paid as fees of the 440 transactions which were included in the block.

Details

Hash	0x051-f1e9a	Mined	7/14/2026, 11:37:59
Parent Hash	0x244-4dae1	Miner	0x48-5f97
Sha3Uncles	0x1dc-49347	Transactions	440
State Root	0xbd8-57ca5	Internal Txs	818
Nonce	0	Sent	64,282776
Depth	5		114,708 USD
Capacity	16.18%	Internal Value	\$114,708
Distance	0m 58s	Value Today	\$114,605
Uncles	0	Average Value	0.14610 ETH
Uncle Reward	0.00000 ETH	Median Value	0.00000 ETH
	0.00 USD	Block Reward	0.00751 ETH
			13.41 USD
Difficulty	0.00000	Mined	0.00000 ETH
Total Difficulty	0.00000		0.00 USD
Gas	40,073,755 66.79%	Fee Reward	0.01963 ETH
Gas Limit	60,000,000		35.03 USD
Size	254,526		

Blockchain



Department of Computer Science & Engineering (IoT)

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

The screenshot shows the Blockchain.com interface. On the left is a sidebar with navigation links: Home, Prices, Charts, SnapMarkets (marked NEW), NFTs, DeFi, Prediction Markets, Academy, News, Developers, and Wallet. Below these are icons for Bitcoin, Ethereum (highlighted), Solana, and Bitcoin Cash, followed by a language selector set to English. The main content area is titled 'Ethereum Block 25,528,989' and notes it was mined on July 14, 2026, at 11:38:23 by an 'Unknown Miner'. A summary paragraph states that a total of 728.6410 ETH (\$1,300,216) were sent in the block, with an average transaction of 6.6848 ETH. It also mentions a total reward of 0.00 ETH \$2.27, consisting of a base reward of 0.00 ETH \$2.27 and an additional 0.0055 ETH (\$9.74) reward paid as fees for the 109 transactions. Below this is a 'Details' section with two columns of data. The first column lists block metadata (Hash, Parent Hash, Sha3Uncles, State Root, Nonce, Depth, Capacity, Distance, Uncles, Uncle Reward, Difficulty, Total Difficulty, Gas, Gas Limit, Size). The second column lists transaction and reward metrics (Mined, Miner, Transactions, Internal Tx, Sent, Internal Value, Value Today, Average Value, Median Value, Block Reward, Minted, Fee Reward). At the bottom, there is a 'Blockchain' section with a right-pointing arrow.

Details			
Hash	0xd7d-2d4ed	Mined	7/14/2026, 11:38:23
Parent Hash	0x5c1-c4896	Miner	0xba-7ed1
Sha3Uncles	0x1dc-49347	Transactions	109
State Root	0x7b6-fdbcb	Internal Tx	27
Nonce	0	Sent	728.640967
Depth	6		1,300,216 USD
Capacity	12.08%	Internal Value	\$1,300,216
Distance	1m 15s	Value Today	\$1,299,042
Uncles	0	Average Value	6.68478 ETH
Uncle Reward	0.00000 ETH	Median Value	0.00200 ETH
	0.00 USD	Block Reward	0.00127 ETH
Difficulty	0.00000		2.27 USD
Total Difficulty	0.00000	Mined	0.00000 ETH
Gas	13,319,270 22.20%		0.00 USD
Gas Limit	60,000,000	Fee Reward	0.00546 ETH
Size	189,982		9.74 USD

Blockchain technology is transforming the way data and transactions are managed by providing a secure, transparent, and decentralized system. Its ability to ensure data integrity, reduce fraud, and improve efficiency makes it valuable across industries such as finance, healthcare, supply chain, education, and government. As the technology continues to evolve, blockchain is expected to play an increasingly important role in driving digital innovation, enhancing trust, and creating more reliable and efficient systems. Despite challenges such as scalability, regulation, and energy consumption, blockchain has the potential to reshape the future of business and society through its wide range of real-world applications.

Conclusion